

CLASS 9th

BIOLOGY BASIC NOTES

A. Cell Biology

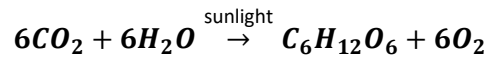
- Cell: Basic unit of life
 - Types:
 - Prokaryotic: No nucleus (Bacteria)
 - Eukaryotic: Nucleus present (Plant & Animal cells)
 - Cell Organelles & Functions:
 - Nucleus: Control center
 - Mitochondria: Powerhouse, ATP production
 - Chloroplast: Photosynthesis
 - Ribosome: Protein synthesis
 - Endoplasmic Reticulum (ER): Transport, rough ER (protein), smooth ER (lipid)
 - Golgi apparatus: Packaging & transport
 - Lysosome: Digestive enzymes
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B. Tissues

- Plant Tissues:
 - Meristematic: Growth (Apical, Lateral)
 - Permanent: Simple (Parenchyma, Collenchyma, Sclerenchyma), Complex (Xylem, Phloem)
 - Animal Tissues:
 - Epithelial: Covering
 - Connective: Support (Bone, Cartilage, Blood)
 - Muscular: Movement (Skeletal, Cardiac, Smooth)
 - Nervous: Control (Neurons)
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C. Nutrition

- Types: Autotrophic (Plants) vs Heterotrophic (Animals, Fungi)
- Photosynthesis:



- Human Digestive System: Mouth → Esophagus → Stomach → Small Intestine → Large Intestine → Rectum
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D. Respiration

- Aerobic: Glucose + O₂ → CO₂ + H₂O + Energy
 - Anaerobic: Glucose → Lactic acid + Energy (muscles), Glucose → Ethanol + CO₂ + Energy (yeast)
 - Respiration Organs: Lungs, Trachea, Alveoli
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E. Circulatory System

- Heart: 4 chambers (Right Atrium, Right Ventricle, Left Atrium, Left Ventricle)
 - Blood Circulation: Pulmonary & Systemic
 - Blood Components: RBC, WBC, Platelets, Plasma
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F. Excretory System

- Kidneys: Remove urea, maintain water-salt balance
 - Nephron: Functional unit
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G. Reproduction

- Asexual: Binary fission, budding, vegetative propagation
 - Sexual: Fusion of gametes
 - Human Reproductive System: Male (Testes → Sperm), Female (Ovaries → Egg)
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H. Genetics & Evolution

- Mendelian Genetics: Dominant & Recessive traits
- DNA & Chromosomes: DNA → Genes → Chromosomes
- Evolution: Natural selection, adaptation